

SPECIAL EMBRYOLOGY*Musculoskeletal - Neuroanatomy***M.C.Q's****Choose only one answer (most appropriate):**

- 1- Regarding the development of vertebrae and intervertebral discs, mark the correct statement:**
 - a- Cleft vertebra (spina bifida) results from non union of vertebral bodies.
 - b- The notochord forms nucleus pulposus of the intervertebral disc.
 - c- The body of the vertebra is segmental in origin.
 - d- The costal process of the vertebra gives rise to the body of vertebra in the lumbar region.
- 2- Regarding the development of the skull, mark the correct statement:**
 - a- The sides and roof of the skull develop by membranous ossification.
 - b- Bones of the face are derived from the sclerotomes of the somites.
 - c- The base of the skull develops by membranous ossification.
 - d- Scaphocephaly results from early closure of the coronal suture.
- 3- Regarding the development of the skull, mark the correct statement:**
 - a- Hypophyseal cartilage forms the body of the maxilla.
 - b- Trabeculae cranii forms the sphenoid bone.
 - c- In microcephaly, the skull fails to expand.
 - d- The occipital bone is derived from neural crest cells.
- 4- Regarding the congenital abnormalities of limbs, mark the correct statement:**
 - a- Absence of third finger and third metacarpal bone is called Lobster claw hand deformity.
 - b- Amelia is an abnormality in which there is an extra finger.
 - c- Congenital hip dislocation is more common in males.
 - d- Syndactyly is an abnormal presence of extra toe.
- 5- Regarding the pharyngeal arches, mark the correct statement:**
 - a- The second arch consists of maxillary and mandibular prominences.
 - b- The facial nerve supplies the derivatives of the third arch.
 - c- The muscles of facial expressions are derived from the second arch.
 - d- The incus and malleus are derived from the third arch.

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- 6- The following structure is derived from the first pharyngeal arch:**
- a- Stapes.
 - b- Meckel's cartilage.
 - c- Muscles of the face.
 - d- Posterior belly of digastric muscle.
- 7- The first pharyngeal arch gives the following structure:**
- a- Orbicularis oris muscle.
 - b- The posterior belly of digastric muscle
 - c- Stapes.
 - d- Masseter muscle.
- 8- Which one of the following structures originates from the second pharyngeal arch?**
- a- Trapezius muscle.
 - b- Temporalis muscle.
 - c- Occipitofrontalis muscle.
 - d- Tongue muscles.
- 9- The nerve of the second pharyngeal arch is:**
- a- Mandibular nerve.
 - b- Glossopharyngeal nerve.
 - c- Facial nerve.
 - d- Oculomotor nerve.
- 10- The nerve of the third pharyngeal arch is:**
- a- Vagus nerve.
 - b- Glossopharyngeal nerve.
 - c- Mandibular/nerve.
 - d- Maxillary nerve.
- 11- Regarding the pharyngeal pouches and clefts, mark the correct statement:**
- a- The second pharyngeal pouch gives rise to the primordium of the palatine tonsil.
 - b- All pharyngeal pouches are lined with ectoderm.
 - c- The fourth pharyngeal pouch differentiates into the inferior parathyroid gland
 - d- The first pharyngeal cleft gives rise to the tubotympanic recess.

12- Regarding the development of the pharyngeal pouches, mark the correct statement:

- a- Tubotympanic recess arises from the 3rd pouch.
- b- Palatine tonsil arises from the first pouch.
- c- Thymus arises from ventral part of 3rd pouch.
- d- Superior parathyroid arises from the dorsal part of third pouch.

13- Regarding the development of the face, mark the correct statement:

- a- The frontonasal process lies caudally in the developing face
- b- The upper lip is formed by fusion of medial and lateral nasal swellings
- c- The median cleft lip results from incomplete fusion of the two lateral nasal swellings.
- d- Unilateral hare lip is produced by failure of maxillary and medial nasal swellings to fuse on one side.

14- Regarding the development of face and nose, mark the correct statement:

- a- The medial nasal swellings form the ala of the nose.
- b- The nasal placodes are mesodermal in origin.
- c- The upper lip is formed by fusion of two maxillary swellings with each other.
- d- The labial component of intermaxillary segment forms the philtrum.

15- Regarding the development of the nose, mark the correct statement:

- a- Oronasal membrane persists throughout period of development of the nasal cavity.
- b- Mucosal diverticula from the medial wall of the developing nasal cavity extend into paranasal sinuses.
- c- Definitive choanae are located caudal to the secondary palate.
- d- Primitive choanae are located caudal to the primary palate.

16- Regarding the development of the palate and nose, mark the correct statement:

- a- The primary palate develops from the mandibular processes.
- b- The secondary palate develops from the frontonasal process.
- c- Posterior partial cleft palate results from failure of fusion of primary and secondary palates.
- d- Oronasal membrane separates the nasal pit from the primitive oral cavity.

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- 17- Regarding the development of the tongue, mark the correct statement:**
- a- The lateral lingual swellings develop from the 3rd pharyngeal arches.
 - b- The hypobranchial eminence develops from the first arch.
 - c- In tongue tie the frenulum linguae usually extends to its tip.
 - d- The foramen cecum lies between the tuberculum impar and the lingual swellings.
- 18- Regarding the development of thyroid gland, mark one correct statement:**
- a- It is derived from the mesoderm of the 1st pharyngeal pouch.
 - b- The thyroglossal duct remains patent after birth.
 - c- During its descent it passes behind the thyroid cartilage.
 - d- It is derived from the floor of the pharynx.
- 19- Regarding the development of tongue and thyroid gland, mark one correct statement:**
- a- The anterior two thirds of tongue are derived from the second and third pharyngeal arch.
 - b- Posterior third of tongue is derived from first arch.
 - c- During development, the thyroid gland remains connected to the tongue by thyroglossal duct.
 - d- The muscles of the tongue are derived from cervical myotomes.
- 20- The following anomaly is related to development of thyroid gland:**
- a- Oblique facial cleft.
 - b- Branchial cyst.
 - c- Dermoid cyst.
 - d- Thyroglossal fistula.
- 21- Regarding the development of the respiratory system, mark one correct statement:**
- a- It appears as an outgrowth from the dorsal wall of the foregut.
 - b- Alveolar epithelial cells produce surfactant substance from the end of the 7th month.
 - c- The visceral pleura is derived from somatic lateral plate mesoderm.
 - d- The cartilage of the trachea is derived from the endoderm.

- N (63) Regarding the development of the central nervous system, mark one correct statement:
- a- It appears as a thickened ectodermal neural plate.
 - b- Prosencephalon gives rise to metencephalon and diencephalon.
 - c- Cervical flexure appears at the junction of hindbrain and spinal cord.
 - d- Myelencephalon will form pons and cerebellum.
- N (64) Regarding the development of the central nervous system, mark one correct statement:
- a- The cephalic part of the neural tube remains narrow.
 - b- Its neuropores establish a temporary connection of the neural tube with the amnion.
 - c- The telencephalon forms pons and cerebellum.
 - d- The myelencephalon forms mid brain.
- N (65) Regarding the development of the brain, mark one correct statement:
- a- Telencephalon gives rise to thalamus and hypothalamus.
 - b- Metencephalon gives rise to medulla oblongata.
 - c- The cavity of the rhombencephalon is known as fourth ventricle.
 - d- The cavity of the mesencephalon is known as lateral ventricle.
- N (66) Regarding the development of the spinal cord, mark one correct statement:
- a- Grey matter of the spinal cord develops in the marginal zone.
 - b- Dorsal root ganglia are derived from the neural crest cells.
 - c- At birth spinal cord ends at the level of the first lumbar vertebra.
 - d- Mantle zone gives rise to the white matter of the spinal cord.
- N (67) Regarding the development of the spinal cord, mark one correct statement:
- a- The dorsal root ganglia are derived from the neural tube.
 - b- At birth, the spinal cord ends at the level of the second sacral vertebra.
 - c- Spina bifida occulta is a condition in which there is failure of the lamina to fuse in three successive vertebrae.
 - d- Meningocele is the condition in which spina bifida is associated with herniation of the meninges.

N (68) Regarding the development of the spinal cord, mark one correct statement:

- a- Basal plate forms the sensory areas of the spinal cord
- b- Alar plate forms the motor areas of the spinal cord
- c- Sulcus limitans marks the boundary between motor and sensory areas.
- d- The roof and floor plates are thick.

69- Regarding the development of the hypophysis cerebri, mark one correct statement:

- a- Its infundibulum gives rise to the posterior lobe.
- b- Rathke's pouch is a downward extension of the diencephalon.
- c- Pars intermedia is derived from the anterior wall of Rathke's pouch.
- d- Pars distalis is derived from the posterior wall of Rathke's pouch.

N (70) Regarding the development of the eye, mark one correct statement:

- a- The optic vesicles are outpocketings of the hind brain.
- b- The intraretinal space disappears completely.
- c- The choroid fissure is present on the upper surface of the optic cup.
- d- The cells of the surface ectoderm in contact with optic vesicle form the lacrimal gland.

N (71) Regarding the development of the eye, mark one correct statement:

- a- The anterior cells of the lens elongate and form lens fibers.
- b- The sclera is ectodermal in origin.
- c- The choroid is mesodermal in origin.
- d- Anophthalmia is too small eye.

N (72) Regarding the developmental anomalies of the eye, mark one correct statement:

- a- Congenital aphakia means no eye development.
- b- Microphthalmia means too small eye.
- c- Congenital cataract means opaque cornea.
- d- Aniridia results from persistence of choroid fissure.

N (73) Regarding the development of ear, mark one correct statement:

- a- Middle ear cavity develops from first pharyngeal cleft.
- b- Malleus develops from mesoderm of second pharyngeal arch.
- c- Ear drum is derived from all three embryonic layers.
- d- Membranous labyrinth is mesodermal in origin.

N 74- Regarding the development of the ear, mark one correct statement:

- a- Otic placodes develop on each side of forebrain.
- b- The ventral component of the otic vesicle form utricle, semicircular canals and endolymphatic duct.
- c- The cochlear duct is separated from the scala vestibuli by basilar membrane.
- d- The semicircular canals appear as flat outpocketing of utricular part of otic vesicle.

N 75- Regarding the development of the ear, mark one correct statement:

- a- The tympanic cavity is ectodermal in origin.
- b- The distal part of the first pouch forms Eustachian tube.
- c- Malleus is derived from the second pharyngeal arch.
- d- The auricle develops from six mesenchymal proliferations of the first and second pharyngeal arches.

76- The diaphragm arises from the following mesodermal structures:

- a- The septum transversum.
- b- The dorsal mesocardium.
- c- The mesentery of the mid gut.
- d- Some lumbar myotomes.

77- The annulus fossa ovalis is a remnant of the:

- a- Truncal septum.
- b- Septum secundum.
- c- Septum primum.
- d- Inter ventricular septum.

78- The fossa ovalis is a remnant of the:

- a- Truncal septum.
- b- Septum primum.
- c- Conus septum
- d- Inter ventricular septum

KEY ANSWERS OF MCQ's

1-	b	27-	b	53-	b
2-	a	28-	c	54-	a
3-	c	29-	c	55-	d
4-	a	30-	b	56-	c
5-	c	31-	d	57-	a
6-	b	32-	d	58-	b
7-	d	33-	d	59-	a
8-	c	34-	c	60-	d
9-	c	35-	a	61-	a
10-	b	36-	b	62-	d
11-	a	37-	b	63-	a, c
12-	c	38-	a	64-	b
13-	d	39-	c	65-	c
14-	d	40-	c	66-	b
15-	a	41-	a	67-	d
16-	d	42-	b	68-	c
17-	c	43-	a	69-	a
18-	d	44-	d	70-	b
19-	c	45-	a	71-	c
20-	d	46-	b	72-	b
21-	b	47-	a	73-	c
22-	b	48-	b	74-	d
23-	d	49-	a	75-	d
24-	a	50-	c	76-	a
25-	d	51-	c	77-	b
26-	c	52-	b	78-	b